Techno-Economic of Photovoltaic Rooftop in Indonesia for Commercial and Residential Customer

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***Abstract*— Despite the high solar energy potential in Indonesia, the number of installed photovoltaic (PV) rooftop is relatively small. To improve the development, the Indonesian government initiate some regulations such as the Ministry of Energy and Mineral Resources (MEMR) policy No. 49/2018 that advertised the 65% incentive and No.16/2019 that remove the cost for parallel capacity and parallel operation for industry customers. However, the number is still far away from the target set by the government. By calculating the techno-economic parameters based on the MEMR tariff regulation No.28/2016 and the Incentive given by the government as per regulation MEMR policy No.49/2018, this paper aims to shows the implementation of the regulation for two different tariff groups such as commercial and residential so that it can be used as a references to improve the policy and PV rooftop development. The result shows that even though the PV rooftop can reduce the electricity cost with saving rate 16.02% for residential and 23% for commercial, the high investment cost with estimate value of IDR 9 M for residential and IDR 360M for commercial, payback in 9 years and a 5.5% return on investment is not so attractive compared to the electricity price from the utility without installing PV. Thus, there should be an improvement on the incentive or policy to increase the number of PV rooftop for commercial and residential tariff group.**

Keywords—PV Rooftop, Techno-Economic, PV Tariff

I. INTRODUCTION

According to the energy plan (RUEN) made by the Indonesian energy council, the government aims to increase the renewable energy share to be 23% [1]. Furthermore, the newest electricity business plan (RUPTL) 2019-2028 state the solar energy target in 2028 will be 908 MW. This includes the 3.2 GW from the PV rooftop. Moreover, the number of PV rooftop is gradually increased since 2013. In 2018, there are 414 customers connected their PV Rooftop to the grid, and in 2019 the number raised to be 609. Although it is increased by 32%, it is still far away compared to the target [2]. Several policies initiated by the government to boost PV Rooftop development. For example, the regulation from Ministry of Energy and Mineral Resources No.49/2018 stated that 65% incentive will be given for the PV Customers and No.16/2019 which declared that the industries customer does not have to pay the cost of parallel capacity and parallel operation.

On the other hand, on-grid PV in China dominated by a large scale than building. Furthermore, significant growth can be seen after 2007. Start from 1.2 MW in 2004, the number increases sharply in 2009 to be 142 MW, wherein 2008 only 21 MW. This rapid change due to the Rooftop Subsidy Program and Golden Sun Demonstration Program which initiated by the government [3]. Also, the electricity tariff for the household customer is \$0.078/kWh [4], while for Feed-in tariff is \$0.025/kWh [5]. It means that the 68% incentive is given to promote the PV Rooftop. For China, the fiscal subsidies proved as the best trigger for PV development [6]. But different case occurs in Thailand. Focus on PV rooftop from 2013 [7], issues like technical and economic became a barrier to its growth [8].

Several PV rooftop study for Indonesia customer has been done, [9] explained the use of PV rooftop in the building for peak load reduction. The investment cost for 117.5 kWp is about IDR2.4 Billion with a payback period of 11 years. Meanwhile, [10] shows the techno-economic calculation for a household customer, where the cost for the PV-battery on-grid system is IDR1,244/kWh and off-grid IDR4,644/kWh. However, both studies were just explaining the techno-economic without point out the incentive regulation and how to calculate it.

Since two of the PV rooftop targets is residential and commercial customers, this paper focuses on the techno-economic of PV rooftop from these customers point of view. There are two scenarios discuss, the first one is the condition of both customers before the PV installed, and the second one where the customers had a mixed source of power from the PV and grid. The technical calculation for energy production and consumption was optimize using HOMER Grid Software, while the tariff calculation is done manually based on regulation No.49/2018 and No.28/2016. Finally, by deducting the bill before and after PV installment, the saving can be calculated and compared.

II. ELECTRICITY TARIFF IN INDONESIA

There are eight different categories of electricity consumers in Indonesia with a total of 38 groups. From this number, 26 groups get subsidies and 12 groups with tariff adjustment. Table I represents different tariff group in Indonesia.

TABLE I. TARIF GROUP IN INDONESIA

Subsidies		
Tariff Group	**Total Group**	**Installed Capacity**
Social	7	S-1/TR 220VA, S-2/TR (450 VA, 900VA, 1300 VA, 2200 VA and 3500VA to 200 kVA), and S-3/TM >200kVA
Residential	3	R-1/TR (450 VA, and 900 VA)
Small Business	4	B-1/TR (450 VA, 900 VA, 1300 VA, and 2200VA to 5500VA)
Small Industries	6	I-1/TR (450 VA, 900 VA, 1300 VA, 2200 VA, 3500 VA to 14 kVA, and I-2/TR>14 kVA to 200 kVA)
Government	4	T/TM >200 kVA
Traction	1	C/TM >200 kVA
Special	6	C/TM >200kVA
Non-Subsidies		
Tariff Group	**Total Group**	**Installed Capacity**
Residential	4	R-1/TR (900VA-RTM, 1300 VA and 2200VA), R-2/TR 3500VA to 5500VA and R-3/TR > 6600VA
Large Business	2	B-2/TR 6600 VA to 200 kVA and B-3/TM >200kVA
Large Industries	2	I-3 /TM >200kVA and I-4/TT > 30000 kVA
Government	3	P-1/TR 6600VA to 200 kVA, P-2/TM > 200 kVA and P-3/TR
Social	1	L/TR, TM, TT

PT.PLN (Persero) responsible for the electricity tariff adjustment (TA) which must be evaluated every three months. However, PT.PLN should report the TA decision and follow the formula given by the Ministry of Energy and Mineral Resources (MEMR). The latest regulation No.28/2016 that already updated four times to be No.3/2020 explained that there are several factors such as exchange rate against the US Dollar, Indonesian crude price (ICP), inflation and coal price were considered as factors that might affect the basic cost of electricity supply. Furthermore, PT.PLN should announce the TA to the customer one month before it enforced. The consumer electricity billing is the sum of load cost (LC) in IDR/kVA/Months, usage cost (UC) in IDR/kWh, and kVArh cost in IDR/KVArh. Except for the household and the business consumers which up to 200 kVA, the kVArh cost (IDR/KVArh) is not applicable. Besides, the residential and commercial customers were divided into small, medium and large groups. From 2015 until 2016, the tariff in Indonesia is gradually decreased with an average of 5-8%. But starting 2016, the tariff for residential and commercial up to 200 kVA is IDR 1467.28 [11] [12].

To calculate the total billing, below equation can be used. The additional cost of public street lighting and tax valid for all the customers in Indonesia where the value depends on the government rules.

$$\text{Total Billing (IDR)} = (\text{Billing (kWh)} \times \text{TA}(\%)) \quad (1)$$

And for the TA (%) calculation, PT.PLN should use the following equation:

$$\text{TA}(\%) = (\text{K}_{\text{Kurs}} \times \Delta\ \text{Kurs}) + \%(\text{K}_{\text{ICP}} \times \Delta_{\text{ICP}}) + \%(\text{K}_{\text{Inflation}} \times \Delta_{\text{Inflation}}) + \%(\text{K}_{\text{Coal}} \times \Delta_{\text{Coal Price}} \quad (2)$$

Where K means coefficient and Δ is a deviation from the new value.

A. Residential

For the residential group with TA, the billing in kWh can is the sum of load cost and usage cost. The load cost can be determined with below equation.

$$\text{LC} = 40\ \text{hrs}\ \times \text{IC (kVA)} + \text{UC (IDR/kWh)} \quad (3)$$

Where the IC is installed capacity and UC is the usage cost.

B. Commercial

For the commercial customers, if the installed capacity is more than 200 kVA, the customer should pay the cost outside the peak load (LWPB) which normally between 23.00-08.00 hrs.

$$\text{LC} = 40\text{hrs} \times \text{IC} \times \text{LWPB} \quad (4)$$

C. PV Rooftop Tariff

The Incentive for PV Rooftop is calculated based on the kWh recorded times 65%. If the customers sell more energy than it consumed, the electricity bill will be cut in the following months [13].

III. ECONOMIC CALCULATION

To see the economy of the overall systems, several parameters like the Levelized cost of energy (LCOE), payback, and return of investment (ROI) should be considered. However, people in general only focus on investment cost and profit. In this case, the monthly and yearly profit can be calculated by subtracting the billing before and after PV installment.

A. Levelized Cost of Energy

Levelized cost of energy is the cost to generate electricity per kWh, which can be calculated with the following equation [14].

$$\text{LCOE} = \frac{C_{\text{ann}}}{E_{\text{serve}}} \quad (5)$$

With LCOE is the Levelized cost of energy (IDR/year), C_{ann} is the annual cost of the system (IDR/year) and E_{serve} is total electricity served (kWh). And to calculate C_{ann}, below equation can be used.

$$C_{\text{ann}} = \text{CRF}(\text{i}, \text{R}_{\text{Proj}}).\, C_{\text{NPC}} \quad (6)$$

Where $\text{CRF}(\text{i}, \text{R}_{\text{Proj}})$ is Capital recovery factor, C_{NPC} is net present cost, i is the annual discount rate (%) and R is project lifetime. In addition, to calculate the real discount rate, the following equation can be used.

$$\text{i} = \frac{\text{i}^1 - \text{f}}{1 + \text{f}} \quad (7)$$

With f is expected inflation rate and i^1 is the nominal discount rate.

IV. Study Case

To see the feasibility of the PV rooftop implementation in the center of the city, it is assumed that there are two different customers at Grogol, Jakarta-Indonesia located at 6°9,7'S, 106°47,6'E with an installed power of 2200 VA for household and 5500 VA for the commercial customer. Furthermore, the usage cost for both customers is IDR 1.467/kWh and the calculation are done without adding the tax and public lighting cost. the optimization of the PV was using HOMER Grid Software. The PV system cost is assumed to be IDR15M/kWp with O&M cost is 3% from cost. The discount rate is assumed to be 10%, with an inflation rate of 3.5% and the project target 20 years.

A. Global Solar Irradiance

The monthly average of the global solar irradiation in this paper was obtained from the NASA database which has been download directly from HOMER grid software. Fig. 1 shows the average of daily radiation (kW/m^2) in the located area.

Fig. 1. Global Solar Irradiance in the located area (kW/m^2)

It can be seen that the GHI on the location is ranging start from 0.17 kW/m^2 to 0.22kW/m^2. It reaches the highest number in September and the lowest in January.

B. Residential

The yearly load for a residential customer is represented in Fig.2. It shows that the average of the load is 11.26kWh/day. Furthermore, Fig.3 represents the daily load data where the highest consumption and the lowest consumption occurred. Although the trend looks different, the average of the people consumed more energy between 17.00-19.00. The highest load occurred on 10/11/2019 at 18.00 hrs with 15kW power rate, while the lowest consumption happened at 16/11/2019 at 0.02 kW at 00.00 hrs

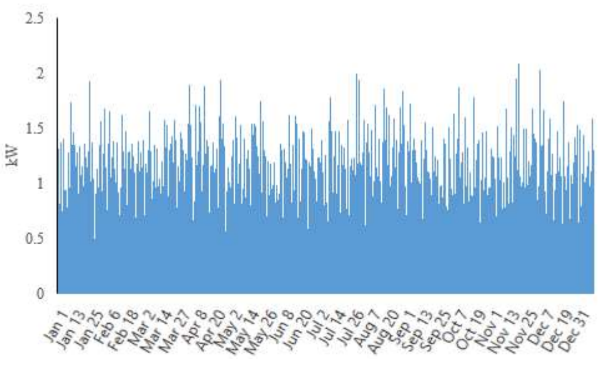

Fig. 2. Residential Load

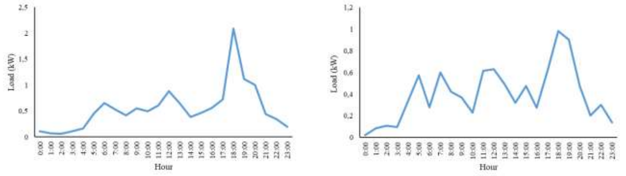

Fig. 3. Residential Load in Different Days

C. Commercial

By Importing the yearly load data per hour to HOMER Grid Software, the commercial load consumption and production simulation can be done. Varying between 1 kW until 40 kW, Fig.4 shows the commercial load where the highest consumption of 40 kW occurred in 19/08/2019 at 15.00 hrs. and the lowest on 16/11/2019 at 0.00 hrs. The trend can be seen that for the commercial customer in this simulation, the electricity consumption tends to rise from 07.00 to 17.00 hrs.

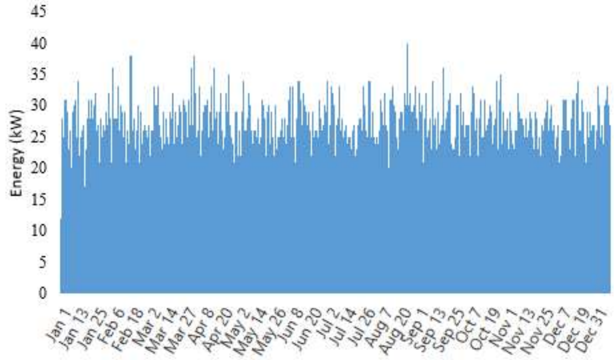

Fig. 4. Commercial Load

Daily load profile might differ from one day to another, this happened because the electricity consumption always depends on user activity. Thus, the number of kWh might different from day to day basis.

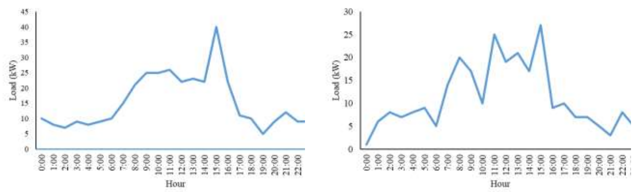

Fig. 5. Commercial Load in Different Days

Fig.5 represents the commercial load on two different days where the highest and the lowest consumption exist.

V. Result and Discussion

The result of the simulation and calculation can be divided into two categories, the technical and economic results. The technical contains the strategy to cover the load by using PV Rooftop and grid, while the economic category shows the comparison of the system with and without PV.

A. Technical Result

To avoid the unmeet load and maximize the PV generation, there should be a dispatch system of electricity supply. The result shows that to cover the residential load with an average of 11.26kWh/day, the ideal PV to be installed is 0.6 kWp. By Installing 0.6 kWp, the electricity production will be 838.35 kWh/year, and the additional supply from the grid will be around 3281.44 kWh/year.

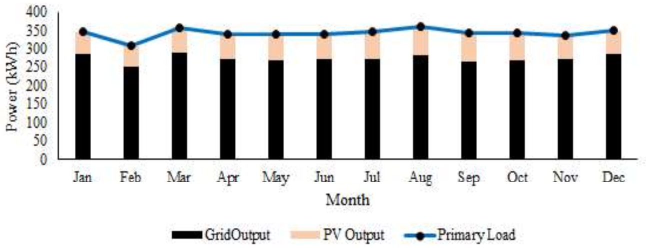

Fig. 6. Residential Production and Consumption

The monthly average of the load ranges from 307.48 kWh in February up to 361.01 kWh in August. As for the PV, the highest production occurred in August with an average of 79.95 kW and the smallest happened in February with 55.91kW. In the peak month of the PV Production, the grid transfer 282.36 kW to cover 361.017 kW load. It means that the average of maximum PV share in the highest production is 22% from the load.

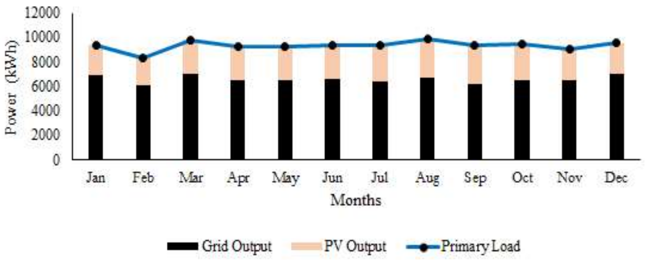

Fig. 7. Commercial Production and Consumption

And for the commercial customer, the suggested PV installed is 24 kW to cover 306.85 kWh/day load with a peak load of 40 kW. The monthly load average is ranging from 9029 kWh in November to 9865 kWh in March. The one-year simulation shows that the PV Production can cover 30% of the load where 70% supplied by the grid.

B. Economic Result

Both residential and commercial customer's calculations show the simple payback will be in 9.08-9.09 years with Return on investment (ROI) about 5.5%. The LCOE for the commercial is IDR 1.302 while for residential is IDR 1.318. With PV installment, the total cost for the electricity consumption varying start from IDR 498K in February up to IDR 548K in December. But, if the supply is only coming from the grid, the electricity bill is ranging from IDR580K in February to IDR640K in December. The saving percentage in each month is different depending on the PV production. The highest saving with PV installment occurred in September with a saving value of 18.26%, while the lowest saving of 13% happened in January. The yearly simulation shows the residential customer with PV installed can save about 16.02% compared to pure grid supply. Fig.8 represents the saving percentage, the cost of electricity consumption with PV and without PV.

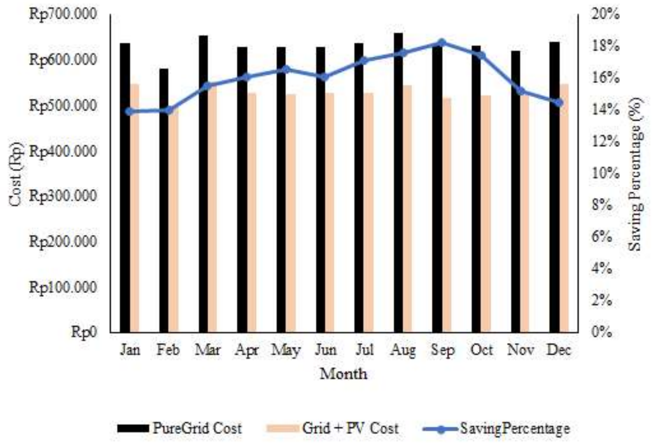

Fig. 8. Saving Percentage from Grid and Combine System Residential Customer

Different consumer behavior will create a different pattern on energy consumption. Thus, the electricity supply also diverse. Without the PV, the commercial customer should pay around IDR 11M up to IDR 13M per month. But, when the system is combined with PV, the monthly electricity bill is decreasing start from 23% to 38%. Yearly, commercial customers will able to save about 23%.

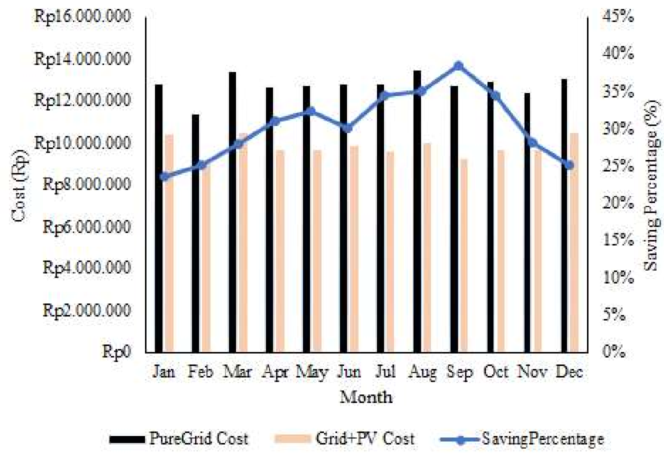

Fig. 9. Saving Percentage from Grid and Combine System Residential Customer

VI. CONCLUSION

From the technical point of view, the important thing is to cover the load and avoid the unmeet load. This could be done by combining the PV with the grid. The result shows that the best size of PV to be installed is 0.6 kWp for the residential customer (2200VA) and 24 kWp for the commercial customer (5500VA). By having a PV as another source of energy, the power consumption from the grid can be reduced up to 30%. And from the economic point of view, the critical point is the saving percentage, payback period and investment cost. It reveals that by investing IDR 9M for residential and IDR 360M for commercial, the saving percentage is up to 23% per year with a payback period of 9 years and a 5.5% return of investment. Though the saving percentage is quite huge, compared to pure grid connection the payback period and ROI might not enough to attract people to use PV Rooftop. Especially with the investment cost which not so cheap. Thus, the government should evaluate the policy to attract more people in the residential and commercial tariff groups. The future works of this study is to investigate the implementation of the PV rooftop policy in other tariff group, and to discuss in detailed the overall situation in order to propose a better solution to increase the number of PV rooftop in Indonesia.

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